AI-Driven DTGEP: An Agentic Methodology for Software Pre-Project Decision-Making

Adrián Fernández Galán, **María Isabel Sánchez Segura**, **Fuensanta Medina Domínguez**

Universidad Carlos III de Madrid (UC3M), Leganés, Madrid, Spain

Abstract

Nearly 70% of digital transformation initiatives fail due to flawed technological decisions in the pre-project phase and subsequent technical debt accumulation. While frameworks like SWEBOK formalize engineering decision-making, organizations systematically bypass this analysis due to the prohibitive time and cost of manually eliciting intangible assets. To eliminate this bottleneck, we present *AI-Driven DTGEP*, a methodology that integrates conversational AI agents into pre-project decision-making. Adapting the DTGEP framework into an asynchronous model, an autonomous agent elicits qualitative data and structures it for deterministic algorithmic execution: maturity assessment (ALTUS), intangible asset and solution deduction (SIPAC), and multicriteria prioritization (AHP). Empirical evaluation across four real-world organizations examined seven language models (from local 4B architectures on consumer hardware to cloud frontier models) alongside a direct-prompting baseline. Results show that agents reliably assess digital maturity with a Mean Absolute Error (MAE) of 1.06–1.27 levels (matching human inter-assessor variability), achieve 88.9% Weighted Recall in identifying candidate technological solutions, and guarantee strict mathematical consistency in prioritization ($\mathrm{CR} < 0.10$, ratios 0.04–0.08). Compared to direct prompting, which hallucinates and violates formal consistency, AI-Driven DTGEP demonstrates that AI can streamline pre-project consulting while preserving strict engineering rigor.

Keywords: Agentic Artificial Intelligence, Engineering Decision-Making, Requirements Elicitation, Software Pre-Project, DTGEP.

1 Introduction

Digital transformation represents a critical strategic necessity for contemporary business competitiveness [1]. Although more than 80% of organizations are actively accelerating their technological initiatives [2], systematic failure rates hover around 70% across industry and academic literature [3], [4]. This failure is not primarily attributable to deficiencies in subsequent software development, but rather to flawed technological decisions during the pre-project phase, triggering a severe accumulation of technical debt that can compromise between 20% and 40% of the total value of an organization's technology estate [5].

As formalized in the SWEBOK v4 guide [6] within its *Software Engineering Economics* knowledge area, software engineering is fundamentally a discipline of choice aimed at aligning technical feasibility with strategic business goals through the *Engineering Decision-Making Process*. The DTGEP (*Digital Transformation Governance Engineering Process*) framework [3] operationalizes this process by decomposing it into structured stages of maturity assessment, organizational asset elicitation, and formal prioritization.

However, the rigorous application of these frameworks faces a fundamental operational barrier: organizations possess a crucial intangible component (tacit knowledge, workflows, human capabilities, and business processes) that serves as the submerged foundation supporting software projects [7]. Multicriteria decision methods such as AHP (*Analytic Hierarchy Process*) [8] require this information to be formally structured. In practice, the elicitation process for these assets lacks automation [6]: it relies entirely on the software engineer through extensive qualitative interviews with stakeholders. Due to the excessive time consumption and high human cost of this manual effort, organizations systematically oversimplify or bypass this stage altogether, selecting technologies without analytical grounding.

To overcome this operational bottleneck, this work introduces *AI-Driven DTGEP*, a methodology that integrates autonomous conversational agents based on Large Language Models (LLMs) under the *ReAct* paradigm [9], [10] into pre-project decision-making. Building upon the DTGEP framework, the proposal adapts its sub-processes into an asynchronous model: the agent relieves the engineer in conducting the interview and capturing information, feeding in a decoupled manner the deterministic execution of formal decision algorithms: digital

maturity diagnosis with ALTUS [11], deduction of intangible assets and technological artifacts with SIPAC [12], and hierarchical multicriteria prioritization using AHP [8].

The primary contributions of this paper are:

1. *Methodological formalization of AI-Driven DTGEP:* A methodology for the software pre-project phase that adapts decision processes into an asynchronous agent-assisted execution model.
2. *Decoupled software architecture:* An agentic engine developed on LangGraph that enforces strict JSON schemas and hierarchical memory to govern the interview and feed deterministic algorithms.
3. *Comprehensive empirical validation and contrast study:* A reproducible benchmark across four real-world organizations evaluating seven language models (open-weight and proprietary), complemented by a comparative study against direct prompting (unstructured LLM inference) that empirically demonstrates the necessity of formal agentic and algorithmic governance.

The remainder of this article is structured as follows: Section 2 reviews related literature and synthesizes the research gap; Section 3 details the methodology and software architecture of AI-Driven DTGEP; Section 4 presents the experimental framework, the research questions, and the empirical findings; Section 5 discusses threats to validity; and Section 6 summarizes our findings and outlines future directions.

2 Related Work

This section reviews the scientific literature at the intersection of software engineering economics in the pre-project phase and the emergence of agents based on Large Language Models (LLMs), synthesizing the methodological gap motivating this work.

2.1 Pre-Project Decision Frameworks and Intangible Asset Evaluation

The *Software Engineering Economics* knowledge area formalized in SWEBOK [6] defines the *Engineering Decision-Making Process* as a systematic lifecycle for selecting technological alternatives to maximize business value. However, this cycle remains inherently abstract: it prescribes the formulation, search, and selection stages, but fails to operationalize empirical data capture within real organizations. Pre-project methodological frameworks such as DTGEP [3] demonstrate that rigorous digital transformation planning requires articulating three dimensions: organizational maturity, underlying knowledge assets, and formal alternative prioritization.

To evaluate an organization's baseline state, the literature commonly resorts to maturity models. While established standards like CMMI [13] assess software development process maturity, frameworks such as ITIL [14] and COBIT [15] target IT service management and governance. Similarly, corporate digital maturity models typically evaluate the mere adoption of visible technologies. In contrast, the ALTUS model [11], [16] focuses specifically on knowledge governance, analyzing 51 organizational capabilities across four maturity levels and seven architectural tiers to diagnose which internal capabilities are consolidated before committing to software investments.

Regarding operational activity modeling, traditional frameworks such as *Business Process Management* (BPM) [17], *Value Stream Mapping* (VSM) [18], and enterprise architecture standards like TOGAF [19] represent observable workflows and infrastructure. However, they remain blind to intangible capital (accumulated experience, cultural friction, and tacit knowledge). To bridge this gap, the SIPAC methodology [12], [20] discovers and characterizes process intangible assets by evaluating their business impact and quality level. By projecting these assets onto two-dimensional matrices, they are classified into health states (*Stable*, *Evolving*, *Replaceable*, or *Warning*), allowing the deduction of mitigating technological artifacts targeted at resolving identified operational bottlenecks [7].

In selecting technological investments, industry often relies on financial metrics such as Return on Investment (ROI) or Cost-Benefit Analysis. Nevertheless, these metrics exhibit severe limitations when attempting to incorporate qualitative factors, adoption risks, and strategic alignment. Although Multi-Criteria Decision-Making (MCDM) methods such as TOPSIS [21] and PROMETHEE [22], [23] have been applied in software engineering, the Analytic Hierarchy Process (AHP) [8], [24] provides a distinctive advantage: it hierarchically decomposes conflicting objectives and derives priorities through pairwise comparisons, ensuring formal control over the evaluator's transitive consistency ($\mathrm{CR} < 0.10$).

The DTGEP methodology [3] orchestrates these disciplines by sequentially integrating ALTUS (maturity diagnosis), SIPAC (intangible asset characterization and artifact deduction), and AHP (multicriteria prioritization). However, all these methodologies share a critical operational barrier: they assume that organizational data are already formalized. In pre-project practice, knowledge regarding capabilities and processes resides tacitly in the minds of stakeholders [20]. Manual qualitative elicitation through semi-structured interviews [6] renders the software engineer an unsustainable operational bottleneck, causing organizations to bypass these methodologies and adopt technology without analytical justification.

2.2 AI in Intangibles and LLM Agents in Software Engineering

Prior work has explored applying Artificial Intelligence to intangible asset modeling. Recent studies combine Natural Language Processing (NLP) and process mining (the C4PM method) to discover intellectual capital from document repositories and system logs [25]. Furthermore, simulations based on System Dynamics and Agent-Based Modeling have been proposed to study process asset behavior [26], [27]. Nonetheless, these approaches share a common limitation: they are passive, *post-hoc* analytical methodologies dependent on pre-existing digital records. In early pre-project phases, tacit knowledge is not recorded in any repository, requiring proactive conversational elicitation directly from human actors.

Concurrently, Large Language Models (LLMs) have disrupted software engineering. A systematic review of 395 primary studies confirms their application across more than 85 software tasks, including code generation, defect detection, and unit testing [28]. In interactive settings, the *ReAct* (*Reasoning and Acting*) paradigm [9] enables models to interleave reasoning with tool execution. Recent proposals such as LLMREI [29] and multi-agent frameworks like MedNegotiator [30] demonstrate that autonomous agents can conduct requirements elicitation interviews with human-level competence. However, these advances are strictly confined to technical requirements engineering for software applications (user stories, functional specifications), without addressing holistic organizational maturity assessment or pre-project intangible asset auditing.

2.3 Limitations of Direct Prompting and Gap Synthesis

Faced with the temptation to replace this methodological framework with direct queries to an LLM (a single prompt requesting an end-to-end digitalization plan), the literature reports critical shortcomings for organizational decision-making. First, LLMs act as statistical manipulators of linguistic form without access to underlying meaning [31], maintaining a persistent risk of hallucinations and ungrounded fabrications regardless of model scale [32]. Second, their self-generated reasoning chains often represent *post-hoc* rationalizations or sycophancy toward user bias rather than formally verifiable logical traces [33]. Furthermore, latent training biases induce premature prescription of commercial platforms [34] without prior evaluation of organizational functional requirements. Finally, there is an intrinsic algebraic inability: unstructured LLMs cannot guarantee mathematical decision properties, such as the reciprocal consistency of pairwise comparisons ($\mathrm{CR} < 0.10$) or deterministic calibration of maturity levels.

Therefore, a dual gap emerges: classical software engineering provides formal rigor and mathematical consistency but lacks automation in qualitative data gathering; generative AI offers adaptive conversational flexibility but lacks axiomatic guarantees, auditability, and consistency. As summarized in Table 1, *AI-Driven DTGEP* addresses this gap by decoupling dialectical interaction (delegated to LLM agents) from decision inference (governed by structured data contracts and deterministic algorithms).

Dimension	Direct LLM Inference (Black-Box)	Decoupled Agentic Architecture (Proposed)
Information Gathering	Single static prompt with free-form text.	Adaptive multi-turn interview with stakeholders.
Organizational Scope	Confined to explicit, pre-existing prompt context.	Systematic elicitation of tacit process knowledge.
Output Structuring	Unstructured narrative text with risk of hallucination.	Strictly typed data schemas validated at runtime.
Decision-Making	Heuristic and stochastic language model inference.	Decoupled execution of formal, deterministic algorithms.
Auditability & Traceability	Opaque reasoning traces and potential sycophancy.	Step-by-step engineer inspection and reproducible models.
Data Governance	Requires sending full operational context to cloud LLMs.	Local model compatibility with strict data isolation.

Table 1: Architectural and methodological contrast between direct LLM inference and a decoupled agentic engineering framework.

3 AI-Driven DTGEP: Methodology and Architecture

The *AI-Driven DTGEP* methodology reconceptualizes the software pre-project phase by structuring the methodological framework into an asynchronous model with a strict separation between conversational qualitative elicitation and deterministic algorithmic processing.

In contrast to traditional consulting, where the human engineer must synchronously guide the interview and manually compute evaluation matrices, AI-Driven DTGEP delegates the elicitation phase to an autonomous AI agent. Once the interaction with the stakeholder concludes, the system immediately and automatically executes algorithmic processing, generating an auditable preliminary draft. Subsequently, the software engineer asynchronously validates the extracted variables and final results, eliminating the temporal concurrency bottleneck (Figure 1).

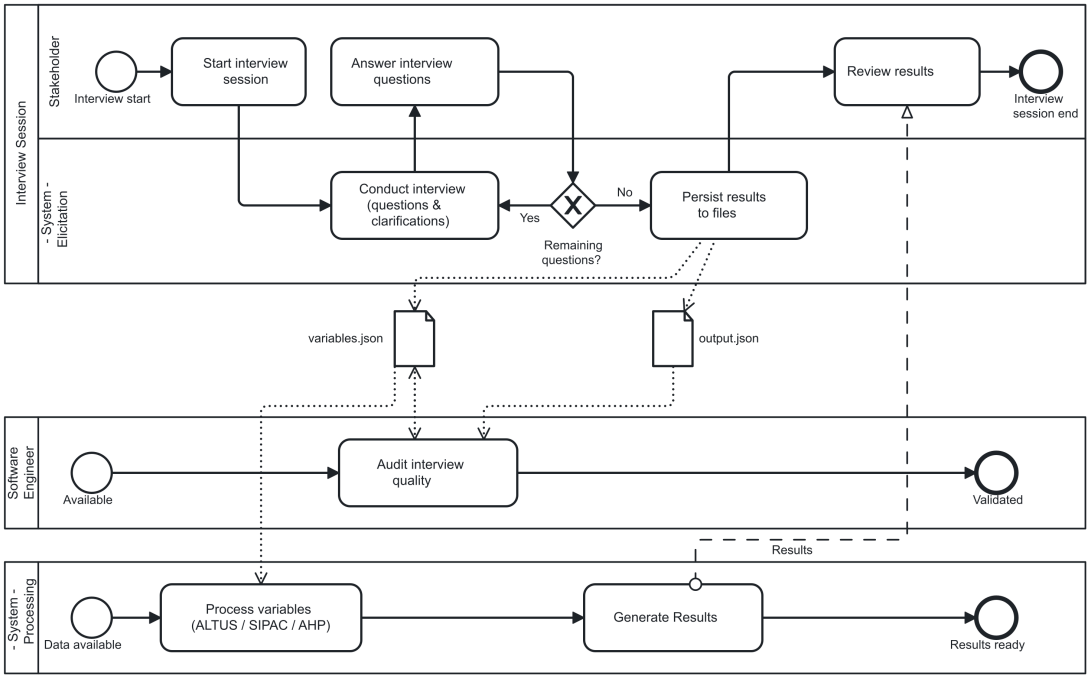


Figure 1: Asynchronous execution model and methodological decoupling between agentic elicitation, deterministic processing, and human audit in AI-Driven DTGEP.

3.1 Agentic Elicitation Engine Architecture

The core of the system is a domain-agnostic conversational engine developed on LangGraph, structured as a deterministic state machine governed by LLMs under the *ReAct* paradigm [9] (Figure 2). The architecture rests upon four foundational pillars:

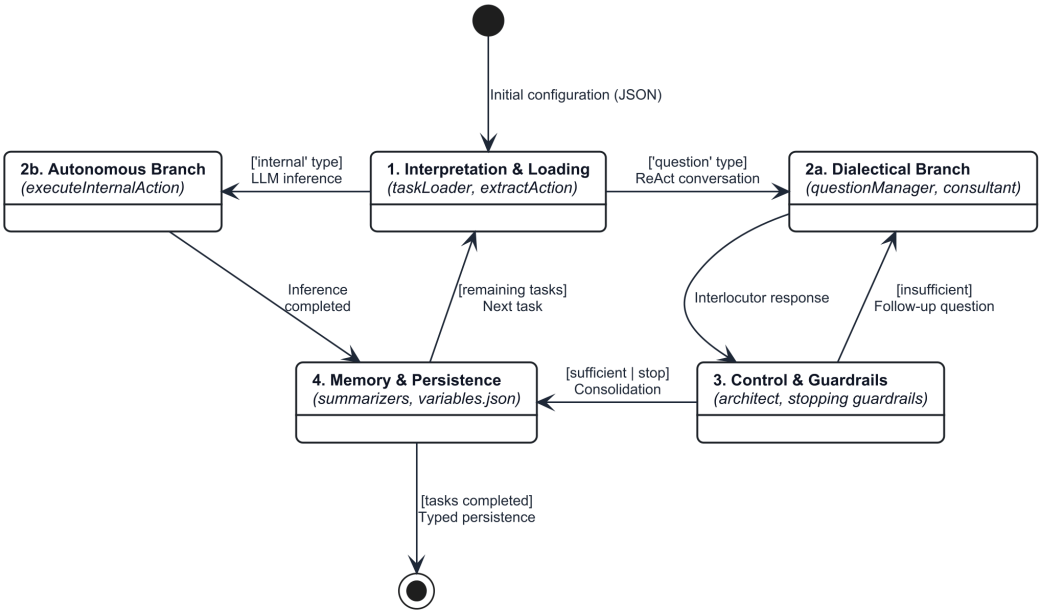


Figure 2: Deterministic LangGraph state machine of the agentic elicitation engine.

1. *Declarative loading and interpretation:* The engine contains no hardcoded logic for any specific process. Its behavior is parameterized via an external JSON configuration contract comprising a hierarchy of tasks and minimal operational actions. Actions are dispatched toward interlocutor interaction (`question`), autonomous internal LLM reasoning (`internal`), or dynamic loop unrolling (`loop_validation`).
2. *Execution bifurcation (dialectical and autonomous branches):* The engine bifurcates flow according to action nature. For interactive actions (`question`), the dialectical branch (`questionManager`) formulates contextualized questions and resolves conceptual queries via `knowledgeConsultant`. For analytical actions (`internal`), the autonomous branch (`executeInternalAction`) reasons directly over context without user intervention, forwarding results to memory without passing through dialogue evaluation.
3. *Sufficiency evaluation and deterministic guardrails:* Operating on the dialectical branch, the `architect` node semantically evaluates whether the user's response provides sufficient evidence. When encountering ambiguous or incomplete answers, the state transitions to a contextualized follow-up loop. To prevent infinite loops and interlocutor fatigue, strict deterministic guardrails are enforced: a cap of 2 clarifications and a ceiling of 3 follow-ups per action, consolidating information once the threshold is reached.
4. *Hierarchical memory and typed persistence:* As actions conclude (interactive or autonomous), the `actionAnswerSummarizer` and `taskSummarizer` nodes condense accumulated context and prune redundant questions in subsequent tasks. Extracted information is formally validated against typed schemas (`outputSchema`) before persisting to structured files (`variables.json`), guaranteeing input integrity for downstream deterministic algorithms.

Building upon this architectural design, *AI-Driven DTGEP* parameterizes the conversational engine via three declarative JSON contracts that sequentially guide elicitation across pre-project stages. First, the *ALTUS Audit* prompts qualitative evidence to score the 51 organizational capabilities of the model (0 to 5), generating `variables_altus.json`. Next, the *SIPAC Characterization* discovers business goals, critical processes, and intangible knowledge assets, deducing the candidate technology artifact catalog and parameterizing their quality and impact indicators in `variables_sipac.json`. Finally, the *AHP Comparison* module consumes these artifacts to formulate pairwise comparisons on Saaty's fundamental scale, consolidating judgment matrices into `variables_ahp.json`.

3.2 Deterministic Algorithmic Decision Processing

Starting from the typed variable files generated by the conversational engine at each stage, the system decouples decision-making by executing three deterministic algorithmic pipelines from the literature, eschewing stochastic weights or LLM biases. First, maturity diagnosis processes `variables_altus.json`, evaluating the 51 capabilities of the ALTUS model [11] to compute aggregated dimension maturity ($M_d = \frac{1}{|C_d|} \sum_{c \in C_d} \text{Level}(c)$, where C_d denotes the set of capabilities belonging to dimension d, and $\text{Level}(c) \in [0, 5]$ is the maturity score of each capability c). Next, the SIPAC pipeline [12] processes indicators in `variables_sipac.json`, computing normalized gap scores toward the target ($V \in [-1, 0]$ for business impact and $W \in [-1, 0]$ for asset quality); this classifies assets and artifacts into four strategic quadrants (*Stable*, *Evolving*, *Replaceable*, and *Warning*) and maps their impact onto the *Business Model Canvas* (BMC). Finally, the AHP module [8] processes pairwise comparison matrices from `variables_ahp.json` via the principal eigenvector equation ($Aw = \lambda_{\max} w$, where A is the reciprocal comparison matrix and w the priority vector) under feasibility, impact, and cost criteria, axiomatically validating the Consistency Ratio ($CR = CI/RI < 0.10$, with consistency index CI and random index RI).

By confining analytical calculation and strategic prioritization to these closed mathematical procedures, the generative AI engine is strictly restricted to conversational elicitation, ensuring that technological prescriptions are auditable, reproducible, and axiomatically consistent.

4 Experimental Evaluation and Results

Since *AI-Driven DTGEP* proposes the agentic operationalization of a standardized pre-project methodology, the experimental evaluation empirically analyzes agent performance across the successive phases of DTGEP, complemented by a qualitative contrast against unprincipled prompting and a runtime viability study. The evaluation is structured around four research questions:

- *RQ1 (ALTUS Phase - Maturity Assessment):* How do agents perform when conducting the ALTUS maturity assessment compared to human consultant evaluations?
- *RQ2 (SIPAC Phase - Solution Deduction):* What is the capability and relevance of agents in identifying intangible assets and deducing technological artifacts in the SIPAC phase compared to the organizational Ground Truth?
- *RQ3 (AHP Phase - Multicriteria Prioritization):* How do agents assist in AHP prioritization, and what degree of qualitative coherence and mathematical consistency do they bring to decision-making?
- *RQ4 (Methodological Contrast against Direct Prompting):* What qualitative and rigor advantages does the structured agentic methodology provide over direct LLM prompting without methodology?

4.1 Experimental Setup and Benchmark

The empirical validation of *AI-Driven DTGEP* was conducted across a benchmark comprising four real-world organizations spanning disparate sectors (specialized e-commerce, herbal distribution, customized textile manufacturing, and renovation/installation services). Derived from original field interviews conducted by human consultants, two reference documentary corpora were compiled: on one hand, the *Organizational Context Dossier*, summarizing each company's factual and operational reality; on the other hand, the formal *Ground Truth*, consisting of the scoring of the 51 ALTUS capabilities (0 to 5), the formal catalog of intangible assets and artifacts deduced in SIPAC, and analytical prioritization matrices in AHP.

To avoid overfitting, pilot case `G1_empresa_a` was exclusively utilized to calibrate the semantic vector filtering threshold ($\tau^* = 0.35$), while final performance was evaluated blindly on the remaining three cases (`G2_empresa_b`, `G3_empresa_c`, `G4_empresa_d`). Seven representative language models were evaluated: three compact local models on consumer hardware (MacBook Air M4, 16 GB unified memory: Gemma 4 E4B, DeepSeek R1 7B, Ornith 9B), three large-scale models served via API (Qwen 3.8 27B, Muse Glimmer 30B, Gemma 4 31B), and one frontier model (Gemini 3.7 Flash ~300B).

4.2 Evaluation of ALTUS Phase: Maturity Assessment (RQ1)

To evaluate the agent's capability to conduct diagnostic interviews against a simulated stakeholder, a conversational stress test was designed: the interlocutor initially provides deliberately terse or ambiguous responses. Faced with this deficit, the agentic engine must detect insufficient evidence and employ follow-up inquiry questions to elicit necessary details before scoring each of the 51 capabilities.

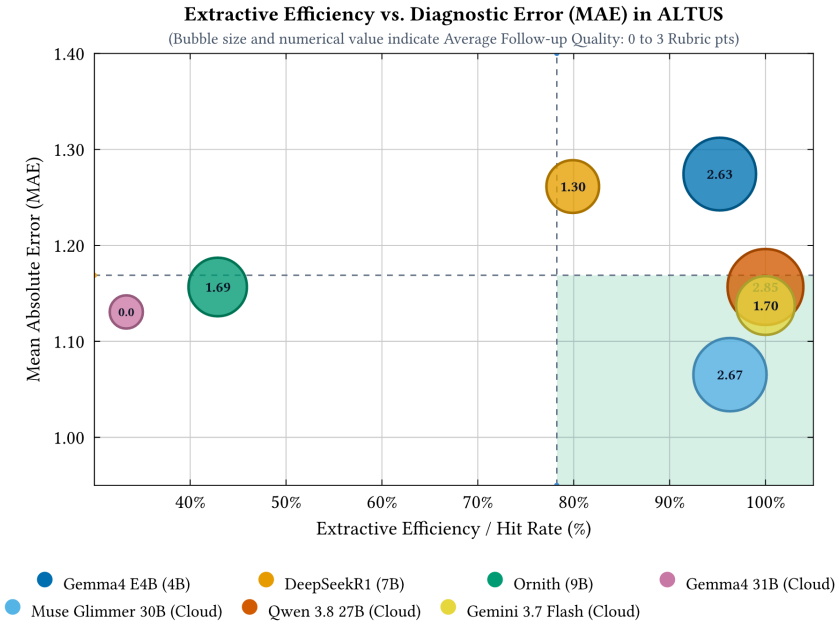


Figure 3: Trade-off between extractive efficacy (Hit Rate, X axis), diagnostic error (MAE, Y axis), and dialectical quality (bubble radius) in ALTUS.

In diagnostic evaluation, exact ordinal agreement (*Accuracy*) ranged between 16.3% and 24.8%, reflecting the rigidity of requiring univocal concordance on qualitative scales. However, Mean Absolute Error (*MAE*) demonstrated a drastic reduction compared to random chance (1.06 to 1.27 points of actual deviation versus 1.94 expected by pure chance, reducing error by 35% to 45%). When considering tolerant accuracy within ± 1 discrepancy level ($\text{Accuracy}_{\pm 1}$), agreement rose to $72\% - 84\%$, falling squarely within the acceptable margin of inter-assessor variability among human consultants [35].

To holistically analyze diagnostic precision alongside conversational dynamics, Figure 3 illustrates the trade-off between extractive efficacy (*Hit Rate*, X axis, mean 88.0%), resulting diagnostic error (*MAE*, Y axis), and follow-up question quality (bubble radius, audited by GPT-5.6 Luna using a graded 0 to 3 rubric). Models such as Qwen 3.8 27B (quality 2.85), Muse Glimmer 30B (2.67), and the local compact Gemma 4 E4B (2.63) converge in the optimal quadrant by uniting high factual extraction with low error, proving they do not predict blindly but through rigorous dialectical inquiry.

4.3 Evaluation of SIPAC Phase: Asset and Artifact Deduction (RQ2)

In SIPAC, the agent generates open-ended natural language proposals structured into process assets and technological artifacts. To isolate the performance of each deductive stage, the evaluation applied *Teacher Forcing*, initializing each stage with validated Ground Truth reference data. Semantic alignment against the Ground Truth was resolved via optimal bipartite matching (Hungarian Algorithm) after applying an embedding cosine similarity pre-filter ($\tau^* = 0.35$). To prevent penalizing valid synonyms, the LLM judge assigned graded equivalence scores (0: discrepancy, $\frac{2}{3}$: partial functional equivalence, 1: exact match), computing weighted metrics (*Weighted Recall* and *Weighted Precision*) over actual solution coverage.

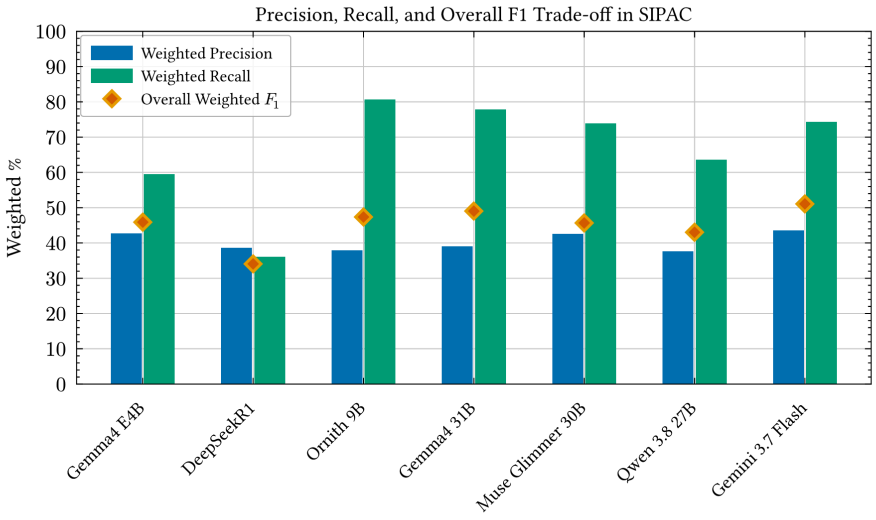


Figure 4: Balance of Weighted Precision, Weighted Recall, and Global F1 in SIPAC across evaluated models.

Figure 4 summarizes metric balance across models. The agents prioritize exhaustiveness over parsimony, achieving a mean *Weighted Recall* of 70.4% that reaches 88.9% for technological artifacts, ensuring that the system comprehensively covers engineering solutions formulated by human consultants without omitting critical pain points.

4.4 Evaluation of AHP Phase: Multicriteria Prioritization (RQ3)

In strategic prioritization, the system combined agentic generation of orthogonal criteria with deterministic resolution of Saaty's matrix.

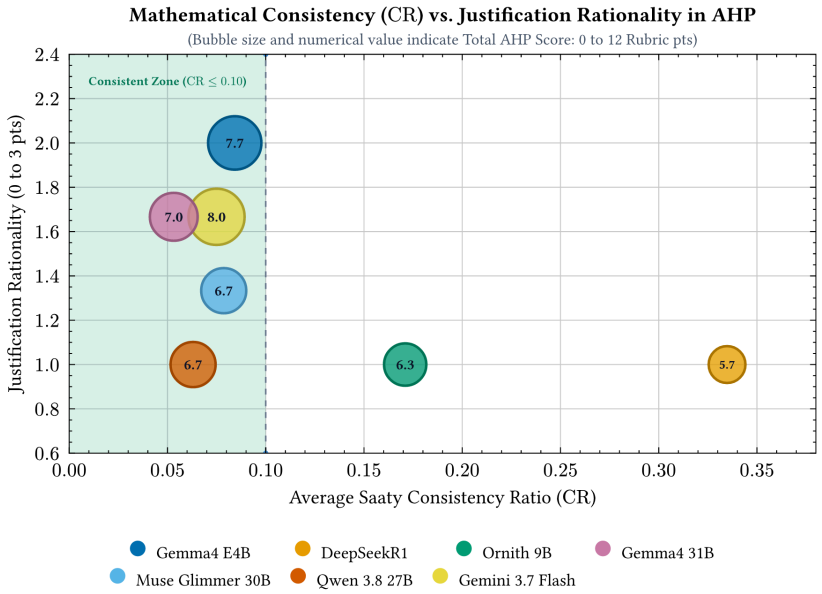


Figure 5: Qualitative quality of generated criteria and analytical consistency in AHP.

As reflected in Figure 5, generated criteria were evaluated by the LLM judge across four dimensions graded from 0 to 3 points (contextual adaptation, orthogonality, holistic completeness, and pairwise justification rationality), with models such as Gemini 3.7 Flash and Gemma 4 E4B achieving average scores exceeding 65% of the overall scale. Algebraically, models with superior contextualization strictly satisfied the transitive consistency constraint imposed by Saaty ($\mathrm{CR} = \frac{\mathrm{CI}}{\mathrm{RI}} < 0.10$, recording mean ratios between 0.05 and 0.08), validating that business-grounded argumentation produces algebraically coherent comparison matrices.

4.5 Qualitative Contrast Against Direct Prompting (RQ4)

To assess the differential value provided by an agentic architecture over unprincipled direct prompting of frontier foundation models, the system was contrasted against a direct prompting baseline. Gemini 3.7 Flash ($T = 0.0$) was supplied with the entire enterprise dossier in a single expert prompt with reasoning, requesting direct generation of the maturity diagnosis, solution catalog, and AHP prioritization.

Dimension	**Direct Prompting Baseline (Gemini 3.7)**	**AI-Driven DTGEP (Proposed)**
Format and Structure	Unstructured Markdown narrative; types unverified.	Strict JSON Schema validated at runtime.
Maturity Assessment	Approximate global scores without per-capability evidence.	Auditable per-capability evaluation (51 ALTUS variables).
Technological Solutions	Premature prescription of commercial SaaS brands (Shopify, HubSpot).	Grounded deduction of custom functional artifacts and KPIs.
AHP Prioritization	Sycophantic weights; mathematically impossible to verify $\mathrm{CR} < 0.10$.	Principal eigenvector resolution with verified consistency ($\mathrm{CR} \in [0.04, 0.08]$).
Intangible Asset Capture	Strictly confined to initial prompt content.	Active elicitation via contextualized follow-up questions.

Table 2: Empirical qualitative and functional comparison between the direct prompting baseline and AI-Driven DTGEP.

The empirical evidence in Table 2 reveals four essential methodological divergences between both paradigms:

In maturity diagnosis (ALTUS), direct prompting acts as a passive recipient: when encountering information gaps, it assumes premises and invents arbitrary ratings without verifying operational reality. In contrast, *AI-Driven DTGEP* continuously monitors factual sufficiency and deploys adaptive follow-up inquiries across the 51 capabilities, guaranteeing an auditable fact-grounded assessment.

In solution deduction (SIPAC), direct prompting incurs *premature commercial solutionism*, prescribing closed third-party SaaS tools (Shopify, HubSpot, Airtable) that induce unjustified early vendor lock-in before requirements are formalized. *AI-Driven DTGEP* deduces vendor-neutral functional specifications guided by business processes and assets.

In multicriteria prioritization (AHP), the baseline assigns sycophantic weights via linear additions, violating criterion orthogonality and precluding transitivity verification. The agentic approach grounds judgments on Saaty's scale and algebraically solves the principal eigenvector, axiomatically guaranteeing the consistency ratio ($\mathrm{CR} < 0.10$).

Finally, in deliverable formalization, direct prompting yields untyped Markdown text unsuitable for automated downstream processing. The agentic architecture persists a typed variable graph in JSON schemas, enabling requirement traceability and seamless export to BPMN models or executive dashboards.

5 Threats to Validity

Construct validity: Evaluating natural language outputs via an LLM-as-a-Judge could introduce evaluator biases. To mitigate this threat, the judge (GPT-5.6 Luna) was completely isolated from the models under test, configured with deterministic temperature ($T = 0.0$), and governed by formally discretized rubrics ($s \in \{0, 1, 2, 3\}$). Furthermore, numerical dimensions (ALTUS maturity scoring and AHP transitive consistency CR) were evaluated using direct mathematical metrics without language model intervention.

Internal validity: The risk of parameter overfitting was neutralized through strict partitioning: pilot case `G1` was reserved to calibrate the semantic threshold ($\tau^* = 0.35$), while the remaining three cases were evaluated blindly. Furthermore, the SIPAC evaluation employed *Teacher Forcing*, isolating each deductive stage with reference data; while this isolates the intrinsic competence of each node without compounding upstream errors, it assumes as a limitation the inability to audit cumulative cascading failure propagation across an end-to-end autonomous run.

External validity and computational viability: Although the benchmark is grounded in four real-world organizations with comprehensive field consulting dossiers, all belong to the small and medium-sized enterprise (SME) segment, introducing a structural bias toward lower digital maturity levels (L_1–L_2). While this accurately mirrors the predominant business landscape, generalization to large multinational corporations requires further validation. Regarding operational viability on consumer hardware without dedicated infrastructure (MacBook Air M4, 16 GB RAM), context accumulation in ALTUS saturated unified memory for models $\geq$ 7B (4.8 hours due to swap degradation), whereas compact 4B architectures (Gemma 4 E4B, 67 unattended minutes) proved to be the optimal configuration for private, sovereign local audits in SMEs without risking sensitive data leakage.

Conclusion validity: The observed inter-model differences and the contrast against direct prompting are supported by closed formal metrics (statistically consistent MAE reduction, $\mathrm{CR} < 0.10$ ratios, and Hungarian algorithm bipartite matches), ensuring the quantitative reproducibility of the findings.

6 Conclusions and Future Work

This article has presented *AI-Driven DTGEP*, a methodology for the software pre-project phase that automates the qualitative elicitation of intangible assets and formalizes technological decision-making through generative agents. By decoupling an asynchronous conversational engine (governed by LangGraph, strict JSON schemas, and hierarchical memory) from the deterministic execution of formal decision algorithms (ALTUS for maturity, SIPAC for solution deduction, and AHP for multicriteria prioritization), the system resolves the historical bottleneck of mandatory synchronous engineer involvement during pre-project requirements gathering.

Empirical validation across four real-world enterprises and seven language models demonstrated that the system achieves a mean maturity error ($\mathrm{MAE} \in [1.06, 1.27]$) comparable to human inter-assessor variability, an 88.9% *Weighted Recall* in discovering technological artifacts, and strict axiomatic consistency ($\mathrm{CR} \in [0.04, 0.08]$ in AHP). Furthermore, the comparative study against a direct prompting baseline proved that unstructured LLM usage lacking an agentic architecture and typed contracts incurs quantitative hallucinations, premature commercial solutionism, and violations of mathematical consistency.

The principal lesson learned is that the true value of AI in software engineering does not lie in outsourcing decisions to a generative black box, but in restricting the model to conversational information elicitation within a rigorous engineering workflow, while reserving quantitative calculation and prioritization for deterministic, auditable algorithms. Additionally, compact open-weight models (4B–9B) running on modest consumer hardware (16 GB RAM) provide sufficient diagnostic competence in ~60 minutes of net execution time, enabling sovereign, privacy-preserving audits in SMEs without risking confidential data exposure.

As priority directions for future work, we envision:

1. Designing asynchronous *multi-stakeholder* interviews to reconcile conflicting perspectives across departments (management, operations, and IT).
2. Incorporating episodic vector memory (selective conversational RAG) to optimize context window consumption during extended sessions.
3. Distilling agentic reasoning into lightweight specialized classifiers to govern graph state transitions in milliseconds, reserving generative LLMs exclusively for open-ended dialectical interaction.

References

[1] OECD, *Measuring the Digital Transformation: A Roadmap for the Future*. Paris: OECD Publishing, 2019. doi: 10.1787/9789264311992-en.

[2] P. Forth, T. Reichert, R. de Laubier, and S. Chakraborty, "Flipping the Odds of Digital Transformation Success." Accessed: Jul. 22, 2026. [Online]. Available: https://www.bcg.com/publications/2020/increasing-odds-of-success-in-digital-transformation

[3] M.-I. Sanchez-Segura, F. Medina-Dominguez, G.-L. Dugarte-Peña, A. de Amescua-Seco, and R. González Cruz, "Engineering sustainable digital transformation projects immune to technical debt," *Kybernetes*, vol. 53, no. 3, pp. 1092–1116, Dec. 2022, doi: 10.1108/K-06-2022-0875.

[4] B. Tabrizi, E. Lam, K. Girard, and V. Irvin, "Digital transformation is not about technology," *Harvard Business Review*, vol. 97, no. 2, p. 13, 2019.

[5] V. Dalal, K. Krishnakanthan, I. Maor, and K.-H. Rathje, "Tech debt: Reclaiming tech equity." Accessed: Jul. 22, 2026. [Online]. Available: https://www.mckinsey.com/capabilities/tech-and-ai/our-insights/tech-debt-reclaiming-tech-equity

[6] R. E. Fairley and others, Eds., *Guide to the Software Engineering Body of Knowledge (SWEBOK Guide)*, Version 4.0. IEEE Computer Society, 2024. [Online]. Available: https://www.computer.org/education/bodies-of-knowledge/software-engineering

[7] M.-I. Sanchez-Segura, G.-L. Dugarte-Peña, A. Amescua-Seco, and F. Medina-Dominguez, "Exploring how the intangible side of an organization impacts its business model," *Kybernetes*, vol. 50, no. 10, pp. 2790–2822, 2021, doi: 10.1108/K-05-2020-0302.

[8] E. H. Forman and S. I. Gass, "The Analytic Hierarchy Process—An Exposition," *Operations Research*, vol. 49, no. 4, pp. 469–486, Aug. 2001, doi: 10.1287/opre.49.4.469.11231.

[9] S. Yao *et al.*, "ReAct: Synergizing Reasoning and Acting in Language Models," in *International Conference on Learning Representations (ICLR)*, 2023.

[10] L. Weng, "LLM Powered Autonomous Agents." Accessed: Jan. 01, 2024. [Online]. Available: https://lilianweng.github.io/posts/2023-06-23-agent/

[11] M.-I. Sanchez-Segura, A. de Amescua Seco, F. Medina-Dominguez, G.-L. Dugarte-Peña, and J.-A. Mora-Soto, "ALTUS: A Process-Oriented, Knowledge Governance Maturity Model," in *Knowledge, People, and Digital Transformation: Approaches for a Sustainable Future*, F. Matos, V. Vairinhos, I. Salavisa, L. Edvinsson, and M. Massaro, Eds., Cham: Springer International Publishing, 2020, pp. 133–162. doi: 10.1007/978-3-030-40390-4_10.

[12] M.-I. Sanchez-Segura, A. Ruiz-Robles, F. Medina-Dominguez, and G.-L. Dugarte-Peña, "Strategic Characterization of Process Assets Based on Asset Quality and Business Impact," *Industrial Management & Data Systems*, vol. 117, no. 8, pp. 1720–1737, 2017, doi: 10.1108/IMDS-10-2016-0422.

[13] M. B. Chrissis, M. Konrad, and S. Shrum, *CMMI for Development: Guidelines for Process Integration and Product Improvement*, 3rd ed. Addison-Wesley Professional, 2011.

[14] AXELOS, *ITIL Foundation: ITIL 4 Edition*. TSO (The Stationery Office), 2019.

[15] ISACA, *COBIT 2019 Framework: Introduction and Methodology*. ISACA, 2019.

[16] M.-I. Sanchez-Segura, F. Medina-Dominguez, A. de Amescua, and G.-L. Dugarte-Peña, "Knowledge governance maturity assessment can help software engineers during the design of business digitalization projects," *Journal of Software: Evolution and Process*, vol. 33, no. 4, p. e2326, 2021, doi: 10.1002/smr.2326.

[17] M. Dumas, M. La Rosa, J. Mendling, and H. A. Reijers, *Fundamentals of Business Process Management*, 2nd ed. Springer, 2018.

[18] M. Rother and J. Shook, *Learning to See: Value Stream Mapping to Add Value and Eliminate Muda*. Lean Enterprise Institute, 1999.

[19] The Open Group, *The TOGAF Standard, Version 9.2*. Van Haren Publishing, 2018.

[20] M.-I. Sánchez-Segura, A. Ruiz-Robles, and F. Medina-Dominguez, "Uncovering hidden process assets: A case study," *Information Systems Frontiers*, vol. 18, pp. 1041–1049, 2016, doi: 10.1007/s10796-016-9622-5.

[21] S. Opricovic and G.-H. Tzeng, "Compromise solution by MCDM methods: A comparative analysis of VIKOR and TOPSIS," *European Journal of Operational Research*, vol. 156, no. 2, pp. 445–455, 2004, doi: https://doi.org/10.1016/S0377-2217(03)00020-1.

[22] J. Brans, P. Vincke, and B. Mareschal, "How to select and how to rank projects: The Promethee method," *European Journal of Operational Research*, vol. 24, no. 2, pp. 228–238, 1986, doi: https://doi.org/10.1016/0377-2217(86)90044-5.

[23] R. Özakıncı and A. Kolukısa Tarhan, "A decision analysis approach for selecting software defect prediction method in the early phases," *Software Quality Journal*, vol. 31, no. 1, pp. 121–177, Sep. 2022, doi: 10.1007/s11219-022-09595-0.

[24] *Uses and Limitations of the AHP Method*. Accessed: Aug. 01, 2026. [Online]. Available: https://link.springer.com/book/10.1007/978-3-030-60392-2

[25] M.-I. Sánchez-Segura, F. Medina-Domínguez, and R. González-Cruz, "Identifying Intangible Assets in Organizations: A Review of Existing Technology-driven Approaches and Proposal of a New One That Improves IC Visibility," in *Intellectual Capital in a Global Business Landscape*, Springer Nature Switzerland AG, 2025, pp. 73–94. doi: 10.1007/978-3-031-86362-2_4.

[26] M.-I. Sanchez-Segura, G.-L. Dugarte-Peña, F. Medina-Dominguez, and C. García de Jesús, "System dynamics and agent-based modelling to represent intangible process assets characterization," *Kybernetes*, vol. 47, no. 2, pp. 289–306, 2018, doi: 10.1108/K-03-2017-0102.

[27] M.-I. Sanchez-Segura, G.-L. Dugarte-Peña, F. Medina-Dominguez, and A. Ruiz-Robles, "A model of biomimetic process assets to simulate their impact on strategic goals," *Information Systems Frontiers*, vol. 19, pp. 1067–1084, 2017, doi: 10.1007/s10796-016-9702-6.

[28] X. Hou *et al.*, "Large Language Models for Software Engineering: A Systematic Literature Review." [Online]. Available: https://arxiv.org/abs/2308.10620

[29] A. Korn, S. Gorsch, and A. Vogelsang, "LLMREI: Automating Requirements Elicitation Interviews with LLMs." [Online]. Available: https://arxiv.org/abs/2507.02564

[30] M. Tanhaei, "MedNegotiator: Automating requirements negotiation in healthcare; simulating clinical and technical perspectives using Generative AI agents," *Array*, vol. 30, p. 100783, 2026, doi: https://doi.org/10.1016/j.array.2026.100783.

[31] E. M. Bender, T. Gebru, A. McMillan-Major, and S. Shmitchell, "On the Dangers of Stochastic Parrots: Can Language Models Be Too Big? 🦜," in *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, in FAccT '21. Virtual Event, Canada: Association for Computing Machinery, 2021, pp. 610–623. doi: 10.1145/3442188.3445922.

[32] Z. Ji *et al.*, "Survey of Hallucination in Natural Language Generation," *ACM Computing Surveys*, vol. 55, no. 12, pp. 1–38, Mar. 2023, doi: 10.1145/3571730.

[33] H. Zhao *et al.*, "Explainability for Large Language Models: A Survey." [Online]. Available: https://arxiv.org/abs/2309.01029

[34] I. O. Gallegos *et al.*, "Bias and Fairness in Large Language Models: A Survey," *Computational Linguistics*, vol. 50, no. 3, pp. 1097–1179, Sep. 2024, doi: 10.1162/coli_a_00524.

[35] K. El Emam, L. Briand, and R. Smith, "Assessor agreement in rating SPICE processes," *Software Process: Improvement and Practice*, vol. 2, no. 4, pp. 291–306, 1996, doi: 10.1002/(SICI)1099-1670(199612)2:4<291::AID-SPIP54>3.0.CO;2-F.